

Markscheme

Mathematics: analysis and approaches
Practice question bank

12. (a)

Given that $2^a = 3$ and $2^b = 5$.

Find the value of 2^{2a-b} .

$$2^{2a-b} = \frac{2^{2a}}{2^b} = \frac{(2^a)^2}{2^b} \quad (M1)$$

substitute the given values (M1)

$$= \frac{3^2}{5} = \frac{9}{5} \quad \text{A1}$$

$$= 1.8 \quad \text{A1}$$

[4 marks]

(b)

Hence find the value of 2^{4a-2b} .

$$2^{4a-2b} = (2^{2a-b})^2 \quad (M1)$$

$$= 1.8^2 = 3.24 \quad \text{A1}$$

[2 marks]